

Two-Stage Optimal Operation Strategy of Isolated Microgrid With TSK Fuzzy Identification of Supply Security

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Abstract—Due to the uncertainty of intermittent energy and system load, it is a big challenge to optimally operate an isolated power system. This article proposes a two-stage optimal operation strategy with a Takagi–Sugeno–Kang (TSK) fuzzy system to address the supply security under uncertainty circumstance. For proper analysis of the uncertainty characteristics, adjustable uncertainty parameters of intermittent energy resource and system load are taken as fuzzy sets; with the consideration of the robustness of these uncertainty parameters on isolated power system, it creates a supply-security identification model with the TSK fuzzy approach under radial basis function (RBF) neural network, and deduces optimal weight values with a recursive least square method. For properly avoiding potential risks, security index is classified into several degrees, each degree of risk can switch a different operation model, which can ensure the supply security of an isolated power system. For properly solving the optimization model, gradient descent-based multiobjective cultural differential evolution is employed to minimize economic cost and emission rate simultaneously. With simulations on isolated regional network, the obtained results reveal that the proposed method can be a viable alternative for optimal operation in isolated power systems.

Index Terms—Intermittent energy, optimal operation, potential risk, Takagi–Sugeno–Kang (TSK) fuzzy approach, uncertainty.

NOMENCLATURE

$\alpha(n)$	Gain parameter of RLS algorithm.
$\alpha_{k0}, \alpha_{k1}, \alpha_{k2}$	Cost coefficients of the k th thermal unit.
ΔT	Time period length.

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ϵ_t
 $\eta_l \in (0, 1]$

$\gamma_{\text{cut},s}$
 $\gamma_{k0}, \gamma_{k1}, \gamma_{k2}$
 $\gamma_{\text{swi},bl}$

$\gamma_{\text{swi},ck}$

κ_j

μ_{ij}

Ω

$\overline{P_{Ij}}$

$\overline{P_{\text{load}}}$

ρ

τ_{store}

τ_{con}

τ_c

τ_i

$\widetilde{P_{Ij}^{[k]}}$

$\widetilde{P_{Ij,\text{max}}}$

$\widetilde{P_{Ij,\text{min}}^{[k]}}$

$\widetilde{P_{Ij,\text{min}}}$

$\widetilde{P_{Ij}}$

$\widetilde{P_{\text{load},\text{max}}^{[k]}}$

$\widetilde{P_{\text{load},\text{max}}}$

$\widetilde{P_{\text{load},\text{min}}^{[k]}}$

$\widetilde{P_{\text{load},\text{min}}}$

$\widetilde{P_{\text{load}}}$

Valve of supply security.

Efficiency factor of the charging or discharging state.

Cost parameter of the cutoff load.

Emission coefficients of the thermal units.

Cost price of switching ON/OFF energy storage.

Cost price of switching ON/OFF thermal units.

Control parameter of RBF function.

Membership function.

Number of power generators.

Estimated output of intermittent energy resource.

Estimated load from the demand side.

Control parameter of RLS algorithm.

Number of energy storage with on state after switching.

Number of controllable load with on state after switching.

Number of thermal units with on state after switching.

Firing degree of rule i .

Upper bounds of the k th small interval of the power output.

Maximum disturbance of intermittent power output.

Lower bounds of the k th small interval of the power output.

Minimum disturbance of intermittent power output.

Output disturbance of intermittent energy resource.

Upper bounds of the k th small interval of the system load.

Maximum disturbance of system load.

Lower bounds of the k th small interval of the system load.

Minimum disturbance of system load.

Load disturbance from the demand side.

$A_{i1}, A_{i2}, \dots, A_{iN}$	Fuzzy sets.
b_{ij}	Width of fuzzy set A_{ij} .
c_j, d_i	Parameters of RBF network.
$f()$	Output function.
H_{bl}	ON/OFF state of the l th energy storage.
H_{bl}	ON/OFF state of the energy storage units.
H_{ck}	ON/OFF state of thermal unit.
$H_{\text{con},s}$	ON/OFF state of s th controllable load.
I	Identity matrix.
k	Current sample index.
K_j	Number of small intervals of intermittent power.
K_{load}	Number of small intervals of load.
m_{ij}	Center of fuzzy set A_{ij} .
N_b	Number of energy storage units.
N_{con}	Number of controllable loads.
N_c	Number of thermal units.
N_R	Number of rules.
p	RLS order.
P_{bl}	Output of the l th energy storage unit.
$P_{ck,\text{max}}$	Maximum output of thermal unit.
$P_{ck,\text{min}}$	Minimum output of thermal unit.
P_{ck}	Output of the k th thermal unit at the t th time period.
$P_{\text{con},s}$	s th a controllable load.
P_{fix}	Fixed part of the system load.
P_{Ij}	Output of intermittent energy resource.
$P_{l,\text{max}}^{\text{cha}}$	Maximum charging output at the $l \in L$ th battery at the t th time period.
$P_{l,\text{max}}^{\text{dis}}$	Maximum discharging output at the $l \in L$ th battery at the t th time period.
P_{load}	System load from the demand side.
P_{loss}	Transmission loss.
P_{store}	Charging/discharging power.
$P_l^{\text{cha}}(t)$	Output of charging state.
$P_l^{\text{dis}}(t)$	Output of discharging states.
$P_{\text{tot}}(t)$	Total output of power generation.
$Q(n)$	Matrix parameter.
r_{Ij}	Control parameters of output uncertainty.
r_{load}	Control parameters of load uncertainty.
S_0	Number set of load.
$T_{ck,t-1}^{\text{off}}$	Off-line time period until the $t - 1$ period.
$T_{ck,t-1}^{\text{on}}$	On-line time period until the $t - 1$ period.
$T_{ck,t-1}^{\text{off}}$	Minimum off-line time period.
$T_{ck,\text{min}}^{\text{on}}$	Minimum on-line time period.
$V_l^{\text{store}}(t)$	Storage of the l th battery at the t th time period.
$V_{l,\text{max}}^{\text{store}}$	Maximum storage of the l th battery.
$V_{l,\text{min}}^{\text{store}}$	Minimum storage of the l th battery.
w_{ij}	Weight of the network.
x_j	j th input variable.
y_i	Output of rule consequent of rule i .
Y_n	Output of the n th sample data.

conditions of power generation and system load [1]. Generally, the objective of optimal operation is to minimize economic cost while satisfying different constraints, including the system load balance, the output limit, spinning reserve constraints, and the ramp rate limit. In [2], emphasis is put on the energy storage system and spinning reserve in the economic dispatch model of microgrids. Also, [3] proposes a distributed economic dispatch strategy for microgrids with multiple energy storage systems, which overcomes the challenges of dynamic couplings among all decision variables and stochastic variables in a centralized dispatching formulation. Besides economic factors, environmental issues can also play a role. In [4], the economic cost and emission rate caused by thermal units are simultaneously optimized, and a set of Pareto optimal schemes is produced. In most instances, the uncertainty or randomness of intermittent energy resources can be considered a tough problem. There are mainly three approaches: 1) fuzzy optimization; 2) stochastic optimization; and 3) robust optimization (RO). The first one depends mainly on the membership function of the decision-makers' experience, which can be subjective and not suitable for real-world applications [5], [6]. Stochastic optimization requires probabilistic information, which is either difficult to obtain or not accurate enough for optimization [7], [8]. RO can deal with the uncertainty problem objectively, as well as, with less probabilistic information, but somehow it may lead to a conservative problem [9], [10]. For avoiding this problem, this article adopts a robustness condition with flexible control parameters, which has been deduced in [11]. The uncertainty of intermittent energy and system load can also lead to a potential risk in isolated power systems. Security problem is also an important issue in the optimal operation of power system, especially in isolated power system. Some researchers focus on the security assessment and evaluation in distributed systems, including distributed power generators [12]–[16]. Usually, the security issue is not taken into consideration at the planning stage, or it exists merely in the communication network. A novel efficient security analysis approach is proposed for overcoming the drawbacks of high computational cost in classical N-k-induced cascading contingency analysis [17]. In [18], approaches are presented for clustering active distribution systems into a set of microgrids with optimized reliability and supply-adequacy indexes. A novel robust security-constrained optimal power flow method is proposed in [19] to balance the economy, combined with the security requirements under the uncertainties associated with renewable generation and load demand. This article adopts supply security to describe the reliability of isolated power systems, and the relationship between supply security and fuzzy sets of power generation and system load is modeled with a Takagi–Sugeno–Kang (TSK) fuzzy system, which has been a hot issue for system identification [20]–[25]. In [20], an interactively recurrent self-evolving fuzzy neural network is proposed for prediction and identification of dynamic systems. A knowledge-leverage-based fuzzy system is discussed from the perspective of transfer learning in [21], which not only make full use of data from the current scene, but also effectively leverage the existing knowledge from reference scenes. In [23], a system identification-based framework is utilized to develop monotone fuzzy IF–THEN rules for formulating monotone zero-

I. INTRODUCTION

WITH the high penetration of renewable energy resources, the great challenge is to adapt the uncertain operating

order TSK fuzzy inference systems. A novel application of a hybrid learning approach is presented in [24] to optimize membership functions of a newly developed interval type-2 intuition fuzzy logic system of TSK fuzzy inference with a neural network learning capacity. This article proposes a two-stage optimal operation strategy with TSK fuzzy identification to optimize the isolated power system; TSK fuzzy identification approach is utilized to identify the relationship between those uncertainty parameters and supply security, and then a two-stage optimization strategy with switching mechanism is presented to minimize economic cost and emission rate simultaneously. The main contributions of this article can be summarized as follows.

- 1) The uncertainty of power generation and system load can lead to the supply-security problem, but the relationship between them cannot be clearly presented. In this article, the uncertainty variables of intermittent energy resources and system load are converted into fuzzy sets, the robustness of these parameters on isolated power system is taken into the identification model, and a TSK fuzzy approach is employed to identify nonlinear relationships between uncertainty variables and supply-security index with recursive least square (RLS) method under an RBF neural network.
- 2) Due to the uncertainty and complexity of isolated power systems, a two-stage framework of optimal operation is proposed to optimize the whole system model, the potential risk of supply security is divided into three different degrees—excellent, good, and bad, each degree can switch a different operation model and improve system security, which can decrease the potential risk to some extent.
- 3) For properly solving those switched optimal operation models, gradient descent-based multiobjective cultural differential evolution (GD-MOCDE) is utilized to optimize the economic cost and emission rate simultaneously. The gradient decent-based mutation operator enhances the convergence ability, which can further improve the optimization efficiency for optimizing the optimal operation model.

The rest of this article is organized as follows. The TSK fuzzy system is presented in Section II and the problem formulation in Section III. The system identification and two-stage optimal operation strategy are proposed in Section IV, and the main results are shown in Section V. Finally, Section VI concludes this article.

II. TSK FUZZY SYSTEM WITH RLS FOR IDENTIFICATION

Generally, a TSK fuzzy system can be considered as a typical nonlinear and dynamic system [26], consisting of several rules that can be expressed as follows:

$$\begin{aligned} \text{Rule } i: & \text{IF}(x_1 \in A_{i1}) \text{AND} \dots \text{AND}(x_N \in A_{iN}) \\ \text{THEN } & y_i = f_i(x_1, \dots, x_N) \quad i = 1, 2, \dots, N_R. \end{aligned} \quad (1)$$

The membership functions are employed to produce membership degree with a Gaussian function as follows:

$$\mu_{ij} = \exp \left[-\frac{(x_j - m_{ij})^2}{b_{ij}^2} \right]. \quad (2)$$

For each rule i , its firing degree τ_i can be considered as “AND” operators for all membership μ_{ij} ($j = 1, 2, \dots, N$), which can obtain

$$\tau_i = \prod_{j=1,2,\dots,N} \mu_{ij} = \exp \left[-\sum_{j=1,2,\dots,N} \frac{(x_j - m_{ij})^2}{b_{ij}^2} \right]. \quad (3)$$

Combined with a weighted average method, it can obtain an output of this fuzzy model as follows:

$$y = \frac{\sum_{i=1}^{N_R} \tau_i y_i}{\sum_{i=1}^{N_R} \tau_i}. \quad (4)$$

Since the output function can be expressed with a nonlinear style, RBF neural network is employed to approximate it with the consideration of two reasons: 1) the RBF has good universal approximation ability and 2) RBF networks are more stable because each individual RBF unit operates only on selected input patterns [27]

$$f_i(X) = \sum_{j=1}^N w_{ij} \phi(\|X - c_j\|) + d_i \quad (5)$$

where $X = (x_1, x_2, \dots, x_N)$, $\phi(\cdot)$ denotes an RBF function, which can be presented with inverse multiquadratics as

$$\phi(\|X - c_j\|) = \frac{1}{\sqrt{\|X - c_j\|_2^2 + \kappa_j^2}}. \quad (6)$$

Since system load can be a dynamic process with online sampling data, a recursive model can be taken into consideration here. For all recent training sample data, it needs to minimize the global objective as follows:

$$J(\theta_1, \theta_2, \dots, \theta_N) = \sum_{n=k-p+1}^k \left(y_n - \sum_{j=1}^N \theta_j \phi(\|X_n - c_j\|) \right)^2 \quad (7)$$

where $X_n = [x_{n-j+1} \ y_{n-j+2} \ \dots \ y_j]^T$ is an input vector of the n th sample data. For simplicity, the abovementioned equation can be rewritten as

$$J(\Theta) = \|Y_n - \Phi\Theta\|_2^2 \quad (8)$$

where the weight vector $\Theta = [\theta_1 \ \theta_2 \ \dots \ \theta_N]^T$, output vector $Y_n = [y_{n-p+1} \ y_{n-p+2} \ \dots \ y_n]^T$, network function vector $\Phi = [\phi]_{n,j}$. To minimize this global objective, the RLS is utilized with several recursive equations as follows:

$$\begin{cases} \Theta(n) = \Theta(n-1) + \alpha(n)G(n) \\ \alpha(n) = Y(n) - \Phi(n)^T \Theta(n-1) \\ G(n) = Q(n-1)\Phi(n) [\rho + \Phi(n)^T Q(n-1)\Phi(n)]^{-1} \\ Q(n) = \rho^{-1}Q(n-1) - G(n)\Phi(n)^T \rho^{-1}Q(n-1) \end{cases} \quad (9)$$

The initial conditions of the above parameters can be set as follows:

$$\begin{cases} \Theta(0) = \mathbf{0} \\ X(n) = X_n = 0, n = -p, -p+1, \dots, -1 \\ Y(n) = y_n = 0, n = -p, -p+1, \dots, -1 \\ Q(0) = \delta I, \delta \in R \end{cases} \quad (10)$$

Since RLS can be considered as a rolling optimization approach, it can revise the weight value Θ as time goes, which can finally achieve a global optima when the time step ends.

III. PROBLEM FORMULATION: RISK DEGREE-BASED ISOLATED POWER SYSTEM WITH DIFFERENT SWITCHING MODELS

A. Uncertainty Analysis of Intermittent Energy Resources and System Load

Since intermittent energy resources and system load have great uncertainty, it can bring a potential risk for hybrid energy system stability. The intermittent power output and system load can be expressed as follows:

$$\begin{cases} P_{Ij} = \overline{P_{Ij}} + r_{Ij} \widetilde{P_{Ij}} \\ P_{load} = \overline{P_{load}} + r_{load} \widetilde{P_{load}} \\ \widetilde{P_{Ij}} \in [\widetilde{P_{Ij,min}}, \widetilde{P_{Ij,max}}] \\ \widetilde{P_{load}} \in [\widetilde{P_{load,min}}, \widetilde{P_{load,max}}] \\ r_{Ij}, r_{load} \in [0, 1] \end{cases} \quad (11)$$

For further analysis on uncertainty, the uncertainty interval can be divided into several small intervals as

$$[\widetilde{P_{Ij,min}}, \widetilde{P_{Ij,max}}] = \bigcup_{k=1}^{K_j} [\widetilde{P_{Ij,min}^{[k]}}, \widetilde{P_{Ij,max}^{[k]}}] \quad (12)$$

$$[\widetilde{P_{load,min}}, \widetilde{P_{load,max}}] = \bigcup_{k=1}^{K_{load}} [\widetilde{P_{load,min}^{[k]}}, \widetilde{P_{load,max}^{[k]}}] \quad (13)$$

For simplicity, suppose that $K_j = K_{load} = K$, it can be labeled as

$$\begin{cases} x_j = \widetilde{P_{Ij}}, j = 1, 2, \dots, J-1 \\ x_J = \widetilde{P_{load}} \\ A_{jk} = [\widetilde{P_{Ij,min}^{[k]}}, \widetilde{P_{Ij,max}^{[k]}}], j = 1, 2, \dots, J-1 \\ A_{Jk} = [\widetilde{P_{load,min}^{[k]}}, \widetilde{P_{load,max}^{[k]}}] \\ \theta_j = r_{Ij}, j = 1, 2, \dots, J-1 \\ \theta_J = -r_{load} \end{cases} \quad (14)$$

where $x = [x_1, x_2, \dots, x_J]$, $A = [A_{jk}]_{J \times K}$, $\theta = [\theta_1, \theta_2, \dots, \theta_J]$, those small intervals are equally divided, it can obtain

$$\begin{cases} \widetilde{P_{Ij,min}^{[k]}} = P_{Ij,min} + (k-1) \frac{P_{Ij,max} - P_{Ij,min}}{K} \\ \widetilde{P_{Ij,max}^{[k]}} = P_{Ij,max} - k \frac{P_{Ij,max} - P_{Ij,min}}{K} \\ \widetilde{P_{load,min}^{[k]}} = P_{load,min} + (k-1) \frac{P_{load,max} - P_{load,min}}{K} \\ \widetilde{P_{load,max}^{[k]}} = P_{load,max} - k \frac{P_{load,max} - P_{load,min}}{K} \end{cases} \quad (15)$$

B. Economic Dispatch Model With Uncertainty Degree in Isolated Power System

The economic issue is crucial in power systems. It consists of different kinds of economic cost, i.e., power generation cost and benefit, load cutoff cost, ON/OFF cost, and simultaneously, it also produces emission pollution from thermal units, which can be expressed as follows.

- 1) Power generation cost: Generally, the power generation cost can be described with quadratic functions of power output. Here, the ON/OFF state of each thermal unit is also taken into consideration, then it can be expressed as

$$C_{gen} = \sum_{k=1}^{N_c} H_{ck} (\alpha_{k0} + \alpha_{k1} P_{ck} + \alpha_{k2} P_{ck}^2) \quad (16)$$

- 2) Emission volume: Since thermal units can produce emission pollutants during power generation, emission issue can also be taken into consideration, which can be presented as

$$E = \sum_{k=1}^{N_c} H_{ck} (\gamma_{k0} + \gamma_{k1} P_{ck} + \gamma_{k2} P_{ck}^2) \quad (17)$$

- 3) Load cutoff cost: Some system load can be controllable, it can be cutoff to ensure the system balance, but it needs to compensate consumers, which can generate cutoff cost as

$$C_{cut} = \sum_{s \in S_0} \gamma_{cut,s} P_{con,s} \quad (18)$$

$$S_0 = \{s | H_{con,s} = 0\}. \quad (19)$$

If it is ON, $H_{con,s} = 1$; otherwise, $H_{con,s} = 0$.

- 4) Switching ON/OFF cost: The ON/OFF operation of thermal units and energy storage can bring economic cost; the switching cost can be presented as

$$C_{swi,thermal} = \sum_{k=1}^{N_c} \gamma_{swi,ck} |H_{ck}(t) - H_{ck}(t-1)| \quad (20)$$

$$C_{swi,store} = \sum_{k=1}^{N_b} \gamma_{swi,bl} H_{bl} |P_l^{store}| \quad (21)$$

$$C_{swi} = C_{swi,thermal} + C_{swi,store} \quad (22)$$

Its total cost F can be expressed with three parts as

$$F = C_{\text{gen}} + C_{\text{cut}} + C_{\text{swi}}. \quad (23)$$

- 5) Power system transmission loss: In this article, the isolated power system consists of thermal units, intermittent energy sources, and energy storage units. Since different energy resources are widely distributed, there exists a transmission loss among them

$$P_{\text{tot}} = P_{\text{load}} + P_{\text{loss}} = P_{\text{fix}} + \sum_{s=1}^{N_{\text{con}}} H_{\text{con},s} P_{\text{con},s} + P_{\text{loss}} \quad (24)$$

$$P_{\text{loss}} = \sum_{i,j \in \Omega} B_{ij} P_i P_j + \sum_{j \in \Omega} B_{0j} P_j + B_{00} \quad (25)$$

$$P_{\text{tot}} = \sum_{j=1}^{J-1} P_{Ij} + \sum_{k=1}^{N_c} H_{ck} P_{ck} + \sum_{l=1}^{N_b} H_{bl} P_{bl}. \quad (26)$$

- 6) Thermal power generation constraints: During thermal power generation, it needs to satisfy maximum and minimum output limits. With consideration of equipment management issue, each thermal unit cannot be in ON/OFF state permanently, its ON/OFF time must be limited as

$$\begin{cases} P_{ck,\min} \leq P_{ck} \leq P_{ck,\max} \\ \left(T_{ck,t-1}^{\text{on}} - T_{ck,\min}^{\text{on}} \right) (H_{ck}(t-1) - H_{ck}(t)) \geq 0 \\ \left(T_{ck,t-1}^{\text{off}} - T_{ck,\min}^{\text{off}} \right) (H_{ck}(t) - H_{ck}(t-1)) \geq 0 \end{cases} \quad (27)$$

- 7) Energy storage constraints: Energy storage is a supplementary energy resource for intermittent energy, it has storage limits and charging/discharging limits, all these constraints can be presented as follows:

$$\begin{cases} V_l^{\text{store}}(t+1) = V_l^{\text{store}}(t) + P_{bl}(t) * \Delta T \\ P_{bl}(t) = \eta_l P_l^{\text{store}}(t) \\ V_{l,\min}^{\text{store}} \leq V_l^{\text{store}}(t) \leq V_{l,\max}^{\text{store}} \\ P_l^{\text{store}}(t) = P_l^{\text{cha}}(t), \quad \text{if } P_l^{\text{store}}(t) \geq 0 \\ P_l^{\text{store}}(t) = -P_l^{\text{dis}}(t), \quad \text{if } P_l^{\text{store}}(t) < 0 \\ 0 \leq P_l^{\text{dis}}(t) \leq P_{l,\max}^{\text{dis}} \\ 0 \leq P_l^{\text{cha}}(t) \leq P_{l,\max}^{\text{cha}} \\ V_l^{\text{store}}(0) = V_{l,\text{initial}}^{\text{store}} \end{cases} \quad (28)$$

IV. FUZZY SYSTEM-BASED TWO-STAGE OPTIMIZATION STRATEGY

A. TSK Fuzzy System Identification of Supply Security in Isolated Power System

The confidence degree of active power supply security can be considered as an important issue in isolated power system, since it is mainly affected by both power supply and demand uncertainty, but the relationship between these is still uncertain. For properly dealing with this problem, a TSK fuzzy system is employed to build up the relationship, an RBF neural network

approximates the nonlinear function, and the robustness constraint is also satisfied for avoiding potential risk. The confidence degree can be considered as the output of a fuzzy system, which is a function of the adjustable parameter θ_j and input variables with an RBF neural network approach, as

$$y = f(X) = \sum_{j=1}^J \theta_j \phi(||X - c_j||). \quad (29)$$

Combined with the introduced fuzzy system model in Section II, the robustness constraint is also taken into consideration with an RLS approach, those recursive iterations can be updated with several procedures. For proper analysis on uncertainty, the system load balance can be rewritten as

$$\sum_{j=1}^J \theta_j x_j = \overline{P_{\text{load}}} - \sum_{j=1}^{J-1} \overline{P_{Ij}} - \sum_{k=1}^{N_c} P_{ck} - \sum_{l=1}^L P_l^{\text{store}}. \quad (30)$$

Moreover, system robustness can also be taken into consideration for avoiding the worst case, which has been referred to in [11]. Then, the following constraint should be satisfied:

$$\sum_{j=1}^{J-1} \theta_j - \theta_J \leq \delta \quad (31)$$

where $\delta \in [0, J]$ represents the uncertainty degree. Here, it satisfies

$$\delta \geq \sqrt{-2J \ln \xi} \quad (32)$$

where $1 - \xi \in (0, 1]$ denotes the probability of satisfying the constraint requirement. Since the robustness constraint is added into the fuzzy system model, it can be converted into a constrained problem. With due consideration of the correlation among sequences, taking forgetting factors also into consideration, the Lagrange function can be presented as follows:

$$L(\Theta_n) = ||Y_n' - \Phi_n' \Theta_n||_2^2 + \lambda (I_0 \Theta_n - (\delta - \delta') e e^T)^2 \quad (33)$$

where $0 < \zeta < 1$ represents the forgetting factor, $I_0 = [1 \ 1 \ \dots \ 1 \ -1]$, $0 \leq \delta' \leq \delta$, $e = [1 \ 0 \ \dots \ 0 \ 0]^T$. Here, $X_n' = [\zeta^{j-1} x_{n-j+1} \ \zeta^{p-2} x_{n-j+2} \ \dots \ x_n]^T$, and the matrix Φ_n' can also be updated as

$$\begin{pmatrix} \phi'_{11} & \phi'_{12} & \dots & \phi'_{1J} \\ \phi'_{21} & \phi'_{22} & \dots & \phi'_{2J} \\ \vdots & \vdots & \vdots & \vdots \\ \phi'_{p1} & \phi'_{p2} & \dots & \phi'_{pJ} \end{pmatrix}. \quad (34)$$

The RBF of the neural network can be updated with ϕ'_{nj} , which can be rewritten as

$$\phi'_{nj} = \phi(||X_n' - c_j||) = \frac{1}{\sqrt{||X_n' - c_j||_2^2 + \kappa_j^2}}. \quad (35)$$

The relationship between X_n' and $X(n)$ can be expressed as

$$X_n' = B X_n \quad (36)$$

where B can be defined as

$$\begin{pmatrix} \zeta^{j-1} & 0 & 0 & 0 \\ 0 & \zeta^{j-2} & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}. \quad (37)$$

Define an exponentially weighted covariance matrix U_n and vector V_n ; then, it can be expressed as

$$\begin{cases} U_n = X'_n (X'_n)^T = \zeta^2 U_{n-1} + x_n x_n^T \\ V_n = X'_n Y'_n = \zeta V_{n-1} + y_n x_n \end{cases}. \quad (38)$$

For optimizing the abovementioned Lagrange function, a relaxation approach is utilized with Lagrange multipliers. Θ_n can be calculated as

$$\Theta_n = (U_n + \lambda I_0^T I_0)^{-1} (V_n + \lambda(\delta - \delta') I_0^T). \quad (39)$$

To decrease the cumulative error caused by iterations, it needs to update Θ with experienced information

$$I_0 \Theta_n = \delta - \delta'. \quad (40)$$

With those aforementioned iterations, the weight vector Θ_n can be deduced with several samples, which also means that the nonlinear relationship between the confidence degree of the system balance and the uncertainty sets can be approximated.

B. Switching Mechanisms Under Confidence Degree of Supply Security

For proper description of supply security as well as convenience for collecting output value from samples, the confidence degree is defined as

$$\text{Security}_{\text{con}} = \text{Prob} \left(\sum_{i \in N_{\text{node}}} \beta_i |P_{\text{tot},i} - P_{\text{load},i} - P_{\text{loss},i}| < \epsilon \right) \quad (41)$$

where N_{node} represents the number of nodes in isolated power system, β_i is state parameter of i th node, $P_{\text{tot},i}$, $P_{\text{load},i}$, and $P_{\text{loss},i}$ denote total power output, load demand, and transmission loss, respectively. The confidence degree is mainly in the range [0,1], denoting safe when it approaches 1. If the confidence degree cannot be properly satisfied, it needs to take some measures to improve it, energy storage can be turned ON to supplement it, and even cut off some controllable load in the worst case. Since the probability of real-world supply security is unknown, the confidence degree can be easily obtained from samples; confidence degree of each sample can be collected by frequency of same events, and then, it can be taken as the output value of TSK fuzzy model. Moreover, within total power generation, thermal power bears 80% of system load to ensure stable power supply. The confidence degree can be classified at three levels—excellent, good, and bad, and each level will switch ON a different adjusting scheme. An excellent level of confidence degree is in the range [0.9, 1], which means that the system balance has been completely satisfied, then its goal is merely to minimize economic cost. A good level of confidence

Algorithm 1

```

1: procedure S(w)itched adjusting scheme under different
   levels of confidence degree
2:   Evaluate confidence degree  $y$ 
3: Case 1:
4:   if  $y \geq 0.9$  then
5:     Goto optimize ED problem
6:   end if
7: Case 2:
8:   if  $0.8 \leq y < 0.9$  then
9:     Turn on energy storage
10:    For  $l = 1 : L$ 
11:       $H_l = 1$ 
12:       $\delta = \delta - 0.1$ 
13:      Evaluate confidence degree  $y$ 
14:      if  $y \geq 0.9$  then
15:        Goto Case 1
16:      end if
17:    end if
18: Case 3:
19:   if  $y < 0.8$  then
20:     Turn on energy storage
21:     For  $l = 1 : L$ 
22:        $H_l = 1$ 
23:        $\delta = \delta - 0.1$ 
24:       Evaluate confidence degree  $y$ 
25:       if  $y < 0.9$  then
26:         switch off controllable load
27:         For  $s = 1 : N_{\text{con}}$ 
28:            $H_{\text{con},s} = 0$ 
29:            $\delta = \delta - 0.1$ 
30:           Evaluate confidence degree  $y$ 
31:           if  $y \geq 0.9$  then
32:             Goto Case 1
33:           end if
34:         end if
35:       end if
36: end procedure

```

degree is in the range [0.8, 0.9), which indicates that the energy storage should be turned ON to decrease the uncertainty for ensuring system balance. A bad level of confidence degree is mainly in the range [0, 0.8), then some controllable load can also be cut off to the keep system balance. For each time period, the confidence degree must achieve an excellent level, then, it can be optimized. From both a security and economic view, the adjusting procedures can be presented in Algorithm 1.

C. GD-MOCDE Approach for Optimizing the Economic Dispatch Problem of Isolated Power

Combined with the abovementioned procedures, the potential risk has been properly avoided and all the ON/OFF or cutoff states of all power generators and controllable load becomes a certain state, the remaining problem can be considered as a classical economic dispatch problem. Without loss of generality,

the optimal operation model can be converted as follows:

$$\left\{ \begin{array}{l} \min F_1, \min F_2 = E \\ P_{ck, \min} \leq P_{ck} \leq P_{ck, \max} \\ V_l^{\text{store}}(t+1) = V_l^{\text{store}}(t) + P_{bl}(t) * \Delta T \\ V_{l, \min}^{\text{store}} \leq V_l^{\text{store}}(t) \leq V_{l, \max}^{\text{store}} \\ P_l^{\text{store}}(t) = P_l^{\text{cha}}(t), \quad \text{if } P_l^{\text{store}}(t) \geq 0 \\ P_l^{\text{store}}(t) = -P_l^{\text{dis}}(t), \quad \text{if } P_l^{\text{store}}(t) < 0 \\ 0 \leq P_l^{\text{dis}}(t) \leq P_{l, \max}^{\text{dis}} \\ 0 \leq P_l^{\text{cha}}(t) \leq P_{l, \max}^{\text{cha}} \\ V_l^{\text{store}}(0) = V_{l, \text{initial}}^{\text{store}} \\ H_{ck} = 1, H_{bl} = 1, H_{\text{con}, s} = 1, \\ k = 1, 2 \dots \tau_c, l = 1, 2 \dots \tau_{\text{store}}, s = 1, 2 \dots \tau_{\text{con}} \end{array} \right. \quad (42)$$

Here, the GD-MOCDE algorithm is utilized to solve the above problem; the output of the thermal unit and the charging/discharging output of the energy storage units are taken as decision variables. In the framework of the cultural algorithm, a differential evolution strategy promotes the evolution of the population space. In the differential evolution, the mutation operator of DE/rand/1/bin can be expressed as

$$R_{r, G+1} = Q_{r, G} + \beta [(Q_{r1, G} - Q_{r2, G}) + (Q_{r3, G} - Q_{r4, G})] \quad (43)$$

where $Q_{r, G}$, $Q_{r1, G}$, $Q_{r2, G}$, $Q_{r3, G}$, and $Q_{r4, G}$ are selected from the current population set, and it also satisfies $r1 \neq r2 \neq r3 \neq r4 \neq r$; G denotes the generation index, $R_{r, G+1}$ is a trial vector, and β is a mutation parameter. For improving the optimization efficiency, the gradient decent mechanism can be obtained with two spaces Ω^+ and Ω^- , which can be presented as

$$\left\{ \begin{array}{l} \Omega^+ = d \in R^m \mid \nabla g(z)^T d > 0 \\ \Omega^- = d \in R^m \mid \nabla g(z)^T d < 0 \end{array} \right. \quad (44)$$

where $g(z) = [g_1(z), g_2(z), \dots, g_m(z)]^T$ represents the objective function vector, $z = [z_1, z_2, \dots, z_n]$ denotes the variable, d is an arbitrary vector, and $\nabla g(z)$ can be considered as a Jacobi matrix as follows:

$$\begin{pmatrix} \frac{\partial g_1(z)}{\partial z_1} & \frac{\partial g_1(z)}{\partial z_2} & \dots & \frac{\partial g_1(z)}{\partial z_n} \\ \frac{\partial g_2(z)}{\partial z_1} & \frac{\partial g_2(z)}{\partial z_2} & \dots & \frac{\partial g_2(z)}{\partial z_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial g_m(z)}{\partial z_1} & \frac{\partial g_m(z)}{\partial z_2} & \dots & \frac{\partial g_m(z)}{\partial z_n} \end{pmatrix} \quad (45)$$

It also means that the deviation between Q_{G+1} and Q_G can be described as

$$Q_{G+1} - Q_G = -\Upsilon_G \sum_{i \in m} \chi_i \frac{\nabla g_i(z)}{\|\nabla g_i(z)\|} \quad (46)$$

Algorithm 2

```

1: procedure G(D)-MOCDE algorithm for economic
   dispatch in isolated power system
2:   Initialize population set  $Q$  and archive set  $B$ 
3:    $G = 0, B = \emptyset, Size = 0$ 
4:   while  $G < G_{\max}$  do
5:     Check total constraint violation
6:     if  $violation > \epsilon$  then
7:       Embedded constraint handling strategy
8:     end if
9:     GD based mutation operator
10:    Crossover operator
11:    Selection operator
12:    Store non-dominated solutions in archive set  $B$ 
13:     $Size = Size + 1$ 
14:    if  $Size > |B|$  then
15:      Truncate archive set  $B$ 
16:    end if
17:     $G = G + 1$ 
18:  end while
19: end procedure

```

With consideration of a discrete version, if it is a biobjective optimization problem, it can be converted as

$$Q_{G+1}^j = Q_G^j + \psi_1(Q_{r1, G} - Q_{r2, G}) + \psi_2(Q_{r3, G} - Q_{r4, G}) \quad (47)$$

where ψ_1 and ψ_2 are two control parameters, which can be presented as

$$\left\{ \begin{array}{l} \psi_1 = -\frac{\Upsilon_G \chi_1 \text{sgn}(g_1(Q_{r1, G}) - g_1(Q_{r2, G}))}{(Q_{r1, G} - Q_{r2, G})^2 \sqrt{\sum_{j \in n} \frac{1}{(Q_{r1, G} - Q_{r2, G})^2}}} \\ \psi_2 = -\frac{\Upsilon_G \chi_2 \text{sgn}(g_2(Q_{r3, G}) - g_2(Q_{r4, G}))}{(Q_{r3, G} - Q_{r4, G})^2 \sqrt{\sum_{j \in n} \frac{1}{(Q_{r3, G} - Q_{r4, G})^2}}} \\ \Upsilon_G = \Upsilon_0 [(G_{\max} - G + 1)/G_{\max}]^p \end{array} \right. \quad (48)$$

where Υ_G , χ_i , ψ_1 , and ψ_2 are control parameters, $G_{\max}$ denotes the maximum generation number. By replacing the classic mutation operator with the abovementioned gradient descent operator, the convergence speed can be improved. Moreover, since there are some inequality and equality constraints in the model, an embedded constraint-handling technique is employed to deal with them, especially the equality constraints; the details can be found in [28]. The whole procedure of the GD-MOCDE for economic dispatch can be presented in Algorithm 2. In the algorithm, $Size$ represents the current size of the archive set, $|B|$ means the size of the archive set, and $violation$ denotes the total violation of the equality constraints.

V. CASE STUDY

The isolated power system consists of four wind farms, three photovoltaic fields, ten thermal units, and four energy storage units, its structure has been shown in Fig. 1, related details can be found in [29] and [30]. The system load includes a fixed load and a controllable load, which can be cut off when

TABLE I
OUTPUT INTERVAL OF WIND POWER GENERATION

period	wind 1	wind 2	wind 3	wind 4	period	wind 1	wind 2	wind 3	wind 4
00:00-00:59	[32, 45]	[30, 42]	[30, 40]	[25, 34]	12:00-12:59	[16, 22]	[15, 21]	[12, 16]	[13, 19]
01:00-01:59	[35, 45]	[35, 41]	[32, 38]	[29, 35]	13:00-13:59	[20, 26]	[20, 26]	[17, 23]	[17, 23]
02:00-02:59	[35, 44]	[34, 40]	[30, 36]	[25, 43]	14:00-14:59	[25, 31]	[22, 30]	[22, 28]	[21, 27]
03:00-03:59	[29, 35]	[27, 35]	[23, 29]	[18, 24]	15:00-15:59	[30, 38]	[28, 36]	[27, 35]	[25, 33]
04:00-04:59	[20, 28]	[20, 26]	[16, 24]	[12, 18]	16:00-16:59	[26, 34]	[24, 32]	[24, 30]	[22, 28]
05:00-05:59	[15, 21]	[13, 19]	[12, 18]	[10, 18]	17:00-17:59	[24, 30]	[22, 26]	[20, 26]	[19, 25]
06:00-06:59	[18, 26]	[15, 23]	[13, 20]	[13, 20]	18:00-18:59	[22, 28]	[19, 25]	[18, 24]	[17, 23]
07:00-07:59	[22, 28]	[19, 25]	[17, 23]	[14, 22]	19:00-19:59	[15, 20]	[17, 23]	[15, 21]	[15, 21]
08:00-08:59	[22, 30]	[22, 28]	[20, 24]	[17, 23]	20:00-20:59	[22, 28]	[23, 29]	[20, 26]	[19, 25]
09:00-09:59	[20, 26]	[18, 24]	[15, 23]	[15, 21]	21:00-21:59	[25, 32]	[28, 34]	[25, 33]	[23, 29]
10:00-10:59	[17, 23]	[15, 19]	[12, 18]	[12, 18]	22:00-22:59	[31, 39]	[27, 35]	[25, 33]	[23, 30]
11:00-11:59	[17, 23]	[15, 21]	[12, 18]	[12, 16]	23:00-23:59	[33, 43]	[32, 40]	[30, 38]	[27, 35]

TABLE II
OUTPUT INTERVAL OF PV FIELDS

period	PV 1	PV 2	PV 3	period	PV 1	PV 2	PV 3
00:00-00:59	[0, 0]	[0, 0]	[0, 0]	12:00-12:59	[28, 36]	[24, 32]	[26, 34]
01:00-01:59	[0, 0]	[0, 0]	[0, 0]	13:00-13:59	[25, 35]	[23, 29]	[27, 33]
02:00-02:59	[0, 0]	[0, 0]	[0, 0]	14:00-14:59	[23, 29]	[20, 24]	[23, 29]
03:00-03:59	[2, 4]	[0, 0]	[1, 3]	15:00-15:59	[20, 24]	[16, 20]	[20, 24]
04:00-04:59	[4, 6]	[1, 3]	[2, 6]	16:00-16:59	[15, 19]	[14, 18]	[15, 21]
05:00-05:59	[8, 12]	[6, 10]	[7, 11]	17:00-17:59	[10, 14]	[11, 15]	[10, 14]
06:00-06:59	[11, 15]	[10, 14]	[8, 12]	18:00-18:59	[6, 8]	[8, 12]	[6, 10]
07:00-07:59	[15, 21]	[13, 19]	[12, 16]	19:00-19:59	[1, 3]	[3, 5]	[4, 6]
08:00-08:59	[16, 22]	[17, 23]	[17, 23]	20:00-20:59	[0, 0]	[0, 0]	[0, 2]
09:00-09:59	[20, 26]	[20, 26]	[17, 23]	21:00-21:59	[0, 0]	[0, 0]	[0, 0]
10:00-10:59	[23, 29]	[22, 28]	[20, 24]	22:00-22:59	[0, 0]	[0, 0]	[0, 0]
11:00-11:59	[23, 29]	[25, 31]	[24, 30]	23:00-23:59	[0, 0]	[0, 0]	[0, 0]

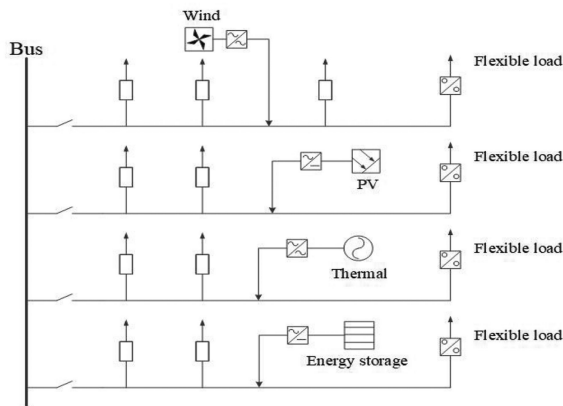


Fig. 1. Structure of isolated power system.

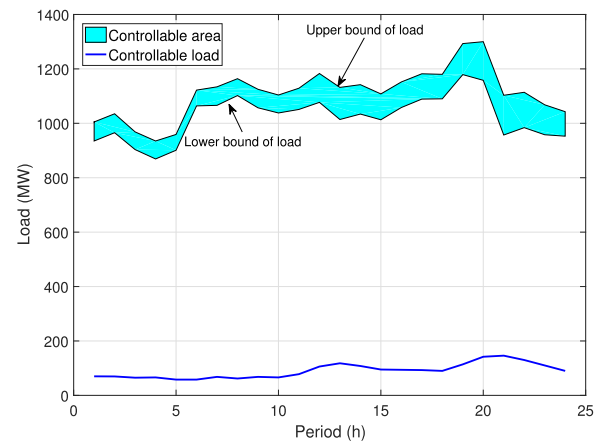


Fig. 2. Controllable load and system load with lower and upper bounds.

necessary. The output of wind power and photovoltaic power can be obtained with wind speed and illumination intensity prediction shown in Tables I and II, system load interval and controllable load are shown in Fig. 2. The outputs of wind

power, solar power, and system load can be considered as fuzzy sets, which can be calculated by dividing the output interval into four parts. Here, four typical periods are selected for verifying the efficiency of both fuzzy identification and

TABLE III
UNCERTAINTY BUDGET OF 24 TIME PERIODS

period	1	2	3	4	5	6	7	8	9	10	11	12
δ	7.342	7.123	6.775	6.642	6.5321	7.221	7.327	7.452	7.4631	7.462	7.4853	7.514
period	13	14	15	16	17	18	19	20	21	22	23	24
δ	7.631	7.642	7.454	7.535	7.498	7.346	7.245	7.524	7.511	7.201	7.132	7.265

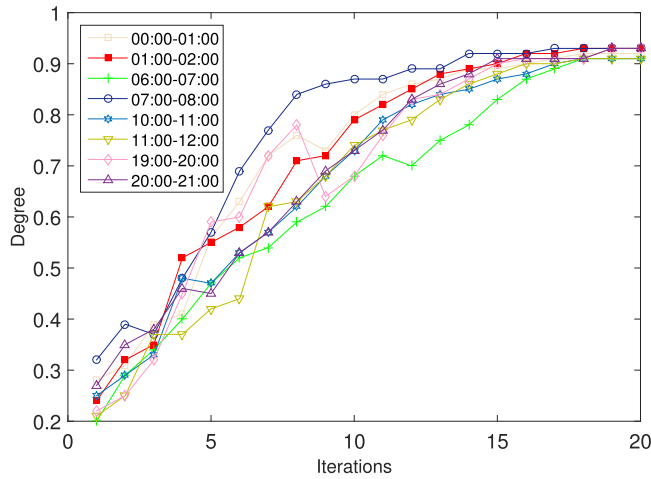


Fig. 3. Convergence analysis of supply security in some typical periods.

optimal operation. It includes 00:00–02:00, 06:00–08:00, 10:00–12:00, and 19:00–21:00, which represent the key periods in one day.

A. Fuzzy System Identification With Intermittent Energy Resources and System Load

The fuzzy system has $4^4 \times 4^3 \times 4 = 65\,536$ possible fuzzy sets and the output value can be calculated with a membership function. Combined with the sampling data, the relationship between the uncertainty variables and supply security can be identified, and then, optimal schemes can be properly applied. A static model can be converted into a dynamic one due to its online identification approach. With the consideration of constraint robustness, the lower bound of the uncertainty budget can be calculated as 6.0697, which also means that the uncertainty budget can only range between [6.0697, 8]. The uncertainty budget for 24 periods are listed in Table III, because the uncertainty is mainly caused by intermittent power generation and system load, the uncertainty budget can be large in some periods where huge intermittent power generation or system load requirement occurs. For each time period, the potential degree of supply security can be evaluated with the TSK fuzzy approach, and then, two-stage optimization strategy is utilized to optimize the isolated power system in comparison with RO. The convergence process of TSK identification in some typical periods are shown in Fig. 3. It can be seen that identification in different typical periods converges in less than 20 iterations to reach an excellent degree (larger than 0.9); the identification is completed after

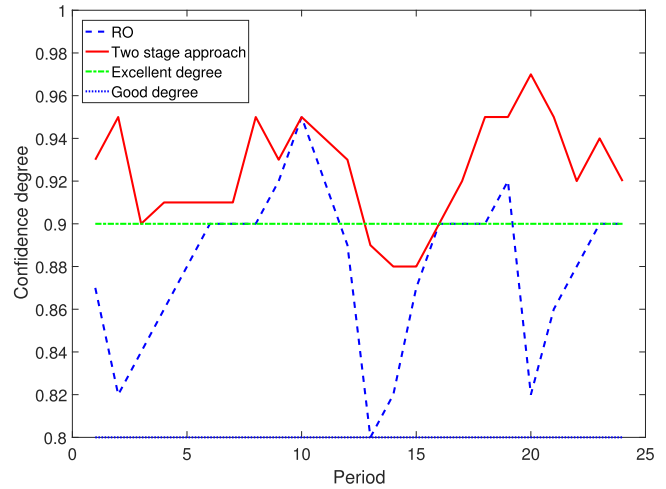


Fig. 4. Confidence degree between RO and two-stage approach.

the switching mechanism, which also means that all the power generators, even system load, work together to improve the supply security. In Fig. 4, the comparison between RO and the proposed two-stage optimization method is presented; it can be seen that the two-stage approach can improve the confidence degree, which can avoid the potential risk of supply security to certain extent.

B. Optimal Operation of an Isolated Power System With a Two-Stage Optimization Strategy

Once the supply security has been identified for each period, optimal schemes can be calculated with a switching mechanism and GD-MOCDE algorithm. Since 00:00–02:00, 06:00–08:00, 10:00–12:00, and 19:00–21:00 can be taken as four typical periods for the system load in one day, the following analysis focuses on the results of these four periods. Since economic cost and emission rate can be taken as two objectives, 20 Pareto schemes are obtained by GD-MOCDE and multiobjective differential evolution (MODE) [31]. The results in some typical periods have been presented in Fig. 5. It can be clearly observed that those Pareto optimal schemes obtained by MODE are disordered, and the GD-MOCDE has both better convergence ability and diversity distribution. For better analysis on optimization results, the tenth scheme of those Pareto optimal schemes is taken as a compromise scheme, which has been labeled in Fig. 5. According to different confidence degrees, different switching mechanisms have been turned ON; the ON/OFF state of all power generators and energy storage units in 24 h are listed in Table IV. It can be seen that Unit 1 and Unit 2

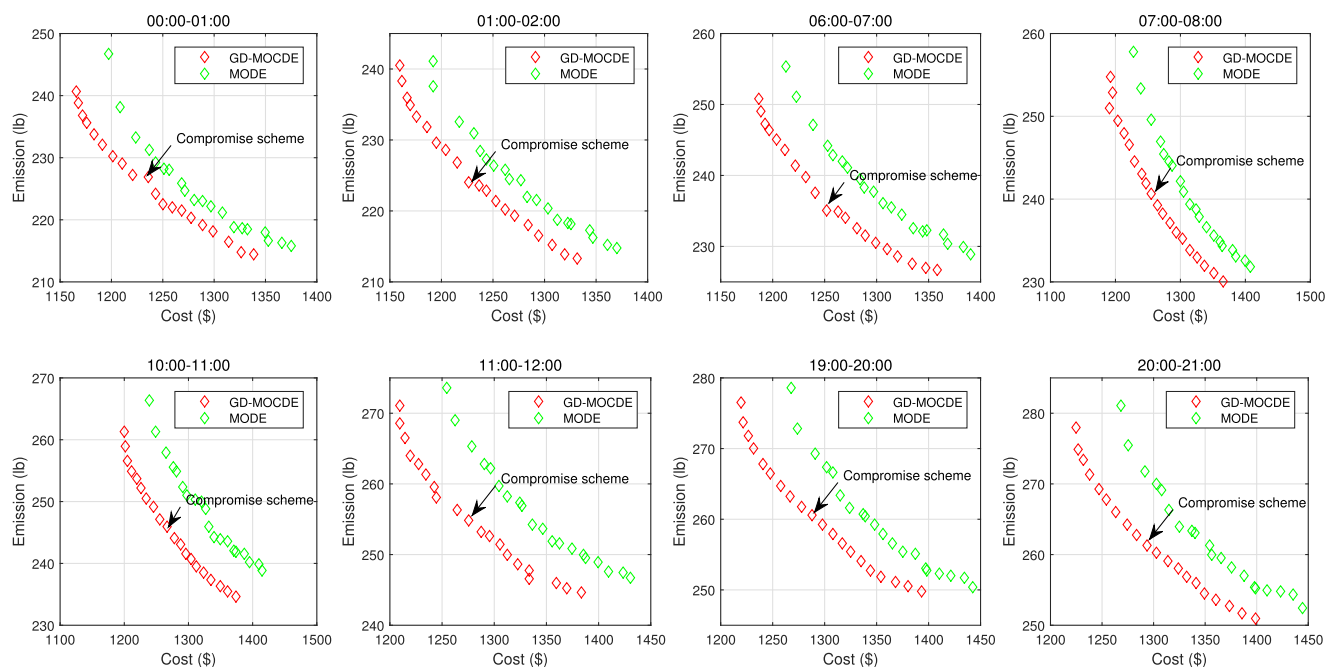


Fig. 5. Pareto optimal schemes by GD-MOCDE and MODE in some typical periods.

TABLE IV
ON/OFF STATE OF POWER GENERATORS AND SYSTEM LOAD

period	Unit1	Unit2	Unit3	Unit4	Unit5	Unit6	Unit7	Unit8	Unit9	Unit10	storage1	storage2	storage3	storage4	load
1	off	off	on	on	on	on	on	on	on	on	on	off	off	off	on
2	off	off	on	on	on	on	on	on	on	on	on	off	off	on	on
3	off	off	on	on	on	on	on	on	on	on	off	off	off	on	on
4	off	off	off	on	on	on	on	on	on	on	off	off	off	off	on
5	off	off	on	on	on	on	on	on	on	on	off	off	on	on	on
6	off	off	on	on	on	on	on	on	on	on	off	off	off	off	on
7	off	off	on	on	on	on	on	on	on	on	on	off	off	off	on
8	off	off	on	on	on	on	on	off	on	on	on	off	off	off	on
9	off	off	on	on	on	on	on	off	on	on	off	off	off	off	on
10	off	off	on	on	on	on	on	off	on	on	off	off	off	off	on
11	on	off	on	on	on	on	on	off	on	on	on	off	off	off	on
12	off	off	on	on	on	on	on	off	on	on	on	off	on	off	on
13	off	off	on	on	on	on	on	off	on	on	on	off	on	off	on
14	off	off	on	on	on	on	on	on	on	on	on	off	on	off	on
15	off	off	on	on	on	on	on	on	on	on	on	off	on	off	on
16	off	off	on	on	on	on	on	on	on	on	on	off	on	off	on
17	off	off	on	on	on	on	on	on	on	on	off	off	off	off	on
18	off	off	on	on	on	on	on	on	on	on	off	off	off	off	on
19	off	off	on	on	on	on	on	on	on	on	off	off	off	off	on
20	off	off	on	on	on	on	on	on	on	on	off	off	off	off	on
21	off	off	on	on	on	on	on	on	on	on	off	off	off	off	on
22	off	off	on	on	on	on	on	on	on	on	on	off	on	off	on
23	off	off	on	on	on	on	on	on	on	on	off	off	on	off	on
24	on	on	on	on	on	on	on	on	on	on	off	off	off	off	on

are almost turned OFF the whole day, energy storage 2 is always turned OFF, the controllable load is not turned OFF, and thermal power is still taken as the main power source. Further analysis is taken on those typical periods, the obtained economic cost and emission rate are presented in Table V, in which the proposed two-stage approach achieves better results than RO.

All the evaluation indexes are presented in Table VI, fuel cost, ON/OFF cost of thermal units, charging/discharging cost, cutoff load cost, total cost, total emission, average confidence degree, and average transmission loss are listed, it can be revealed that the proposed two-stage approach has better results than RO. Since energy storage is seldom used, the switching cost

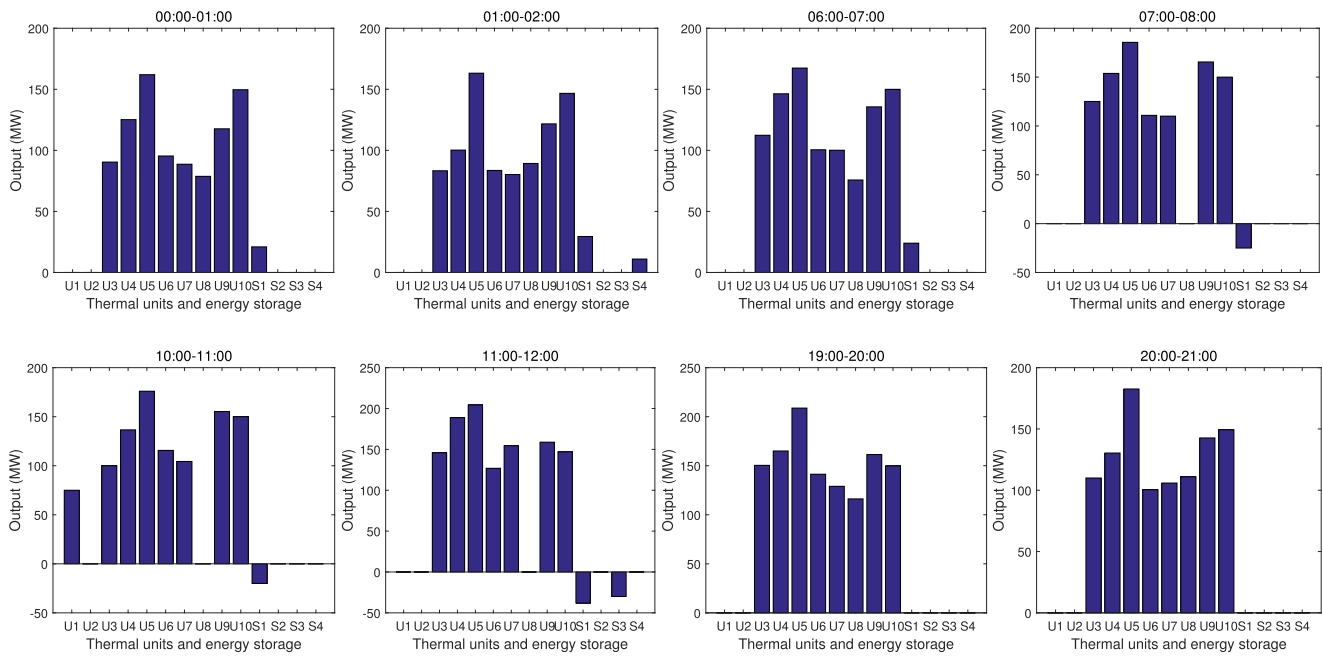


Fig. 6. Output of thermal units and energy storage in some typical periods.

TABLE V
ECONOMIC COST AND EMISSION OF SOME TYPICAL PERIODS

Periods	RO		Two stage approach	
	Cost	Emission	Cost	Emission
00:00-01:00	1780.168	223.142	1535.836	226.8745
01:00-02:00	1776.541	224.348	1526.278	224.077
06:00-07:00	1796.878	237.766	1651.548	235.0249
07:00-08:00	1903.875	239.416	1795.792	240.6667
10:00-11:00	2026.398	248.735	1966.264	245.9567
11:00-12:00	2036.938	254.179	1975.633	254.7754
19:00-20:00	2047.687	259.274	1987.835	260.6285
20:00-21:00	2054.649	261.367	1993.399	261.2676
Total	45733.04	5756.712	43130.62	5766.276

TABLE VI
COMPARISON OF TOTAL ECONOMIC COST (\$), EMISSION RATE (LB),
AND TRANSMISSION LOSS (MW) WITH RO

	RO	Two stage approach
Fuel cost	31623.432	30134.26
On/off cost of thermal unit	5571.96	4812.65
Charging/discharging cost	8537.652	8183.712
Cut-off load cost	0	0
Total cost	45733.044	43130.622
Total emission	5756.712	5766.276
Average confidence degree	0.852	0.915
Average transmission loss	78	66

can be expensive. In those typical periods, the outputs of all power generators and energy storage units are shown in Fig. 6, where U1, U2, ..., U10 represent Unit 1, Unit 2, ..., Unit 10, S1, S2, S3, and S4 represent energy storage 1, energy storage 2, energy storage 3, and energy storage 4. With the

above-mentioned analysis, optimal operation with supply-security identification is a complicated problem. The system model can be uncertain, multiobjective, and complex coupled, hence, this article proposes a two-stage optimization approach with supply-security identification. The TSK fuzzy approach can properly identify the relationship between uncertainty parameters and supply security, and a two-stage optimization strategy can switch ON/OFF power generators to improve the security degree according to three levels of confidence degree, and then, optimize the economic dispatch problem with the known ON/OFF state of the power generators and energy storage units with the GD-MOCDE algorithm; the obtained results can reveal that the proposed method can be an effective way for solving isolated power system problem.

VI. CONCLUSION

The uncertainty and complexity of the optimal operation in isolated power system pose a great challenge optimization. This article proposed a two-stage optimization strategy with a TSK fuzzy identification of supply security. According to the obtained results, the merits can be concluded as follows.

- 1) Supply security can be considered as an important issue in isolated power system, and it is mainly affected by the uncertainty of power generation and system load. For better dealing with the uncertainty problem and decreasing potential risk, this article proposed a TSK fuzzy model for identifying the relationship between uncertainty parameters and confidence degree of supply security with TSK fuzzy model under RBF neural network.
- 2) Due to the complexity-coupled characteristics of the system model, a two-stage optimization strategy was

proposed to optimize the operation model. Combined with different confidence degree of supply security, a switching mechanism was made before economic dispatch to ensure the security degree. It adjusted the ON/OFF state of power generators, energy storage devices, and even cut off system load to keep the balance until the security degree achieved an "excellent" degree.

- 3) After obtaining the ON/OFF state of power generators, energy storage devices, and system load, the GD-MOCDE was utilized to optimize the economic cost and emission rate simultaneously. It searches the optimal scheme with gradient descent directions, which improves the convergence ability in comparison to MODE. With the GD-MOCDE algorithm, 20 Pareto optimal schemes were produced.

These simulation results verified that the proposed two-stage optimization strategy with a TSK identification of supply security can be a viable and promising approach for the optimal operation of an isolated power system.

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